

Prerequisite

You'll need a Kubernetes (K8s) cluster for the homework.

You can use the iac cluster but delete it at the end of the day

Provide the mentors with manifests or commands for your homeworks.

Exercise 1

Create a namespace called *playground* with two custom labels (your choice):

- Query the created resource and check where what's the key associated with the name of your namespace
- Extra: check on the official documentation what *managedFields* stands for

Exercise 2

Create an nginx pod in the namespace created above:

- Use the full path for the image name
- Define an environment variable for the nginx container, can be static (check k8s documentation for additional fields)
- Check inside the container that the value was set
- Extra: use an environment variable as an argument to the nginx container (check k8s documentation for additional fields)

Exercise 3

Look at the events of the above pod and identify what were the steps for that pod to be created (check kubectl events).

Extra Exercise

Create a ServiceAccount *sa-logs* that can check the logs of the pods in *default* and *playground* namespaces only.

- Role and RoleBindings are required

Create a pod using the following docker image <https://hub.docker.com/r/bitnami/kubectl/> and use the above ServiceAccount

- check the logs of a random pod in *default* and *playground* namespace
- check the logs of a random pod in *kube-system*. What happened?